

Syed Nahin Hossain

syednahinhossain@gmail.com

Portfolio: nahin-jzs.github.io/ | Google Scholar : Syed Nahin Hossain | LinkedIn : syed-nahin-hossain-658189176/

Research Interests

3D Vision | Computer Vision | Medical Image Analysis | Computational Imaging | Deep Learning

Education

Bachelor of Science in Computer Science and Engineering

Khulna University of Engineering & Technology, Khulna, Bangladesh

CGPA – 3.50/4.00

Graduated: May 2023

- **Thesis:** Gray Scale to Color Image Conversion using Generative Adversarial Network

Supervisor: Dr. Sk. Md. Masudul Ahsan

Findings:

1. Conditional GAN excels at generating realistic color photos, with training in Lab and YCbCr color spaces proving most efficient for image colorization.
2. Using pre-trained models enhances the generator's efficiency, and focusing on specific tasks yields more efficient colorization.

Research and Publications

Research

1. "Bone Fracture Detection and Localization Using Transfer Learning Techniques on a Comprehensive Dataset."
Status – Manuscript Ready
2. "A Comparative Analysis of GAN-Based Grayscale Image Colorization across Color Spaces, Architectures, and Region-Specific Optimization."
Status – Submitted in Q1 Journal
3. "Advanced Modeling of UHPC Compressive Strength: Insights from Eleven Machine Learning Algorithms."
Status – Manuscript Ready
4. "Monocular 3D Vision for Vehicle Distance Estimation Toward Digital Twin Traffic Systems."
Status – Ongoing work

Journal Publications

1. "Advancing Water Quality Management: Comparative Analysis of Machine Learning Models for Predicting WQI in Dhaka's Rivers." Nishat, M. H., Khan, M. H. R. B., Ahmed, T., **Hossain, S. N.**, Ahsan, A., El-Sergany, M. M., Shafiquzzaman, M., Imteaz, M. A., & Alresheedi, M. T. *Environmental Sciences Europe* (2025). **IF: 6.0** | **Citations: 67** | DOI: 10.1186/s12302-025-01078-w

Conference Publications

1. "Automatic License Plate Recognition System for Bangladeshi Vehicles Using Deep Neural Network." **Hossain, S. N.**, Hassan, M. Z., & Al Masba, M. M. *Proc. Int. Conf. on Big Data, IoT and Machine Learning (BIM)*, Springer (2021). **Best Paper Award** | **Citations: 22** | DOI: 10.1007/978-981-16-6636-0_8
2. "Bangladeshi Local Vehicle Recognition with A Comprehensive Dataset using Transfer Learning Techniques." Rahman, M. S., Hassan, M. Z., **Hossain, S. N.**, Masrur, N., & Rabbi, J. *IEEE* (2022). **Citations: 8** | DOI: 10.1109/STI56238.2022.10103325
3. "Transfer Learning for Vehicle Recognition: A Comprehensive Dataset and Evaluation on Bangladeshi Roads." Prapty, A. S., **Hossain, S. N.**, Efti, M. S. P., Rahman, M. M., & Nurullah, M. *IEEE* (2026). **IEEE Xplore: 11545960**

Work Experience

Lecturer, Department of Computer Science & Engineering

Bangladesh Army University of Science & Technology, Khulna, Bangladesh

July 2025 – Present

Full Time

Lecturer, Department of Computer Science & Engineering

Bangladesh Army University of Engineering & Technology, Natore, Bangladesh

July 2023 – July 2025

Full Time

Technical Projects

Automatic License Plate Recognizer	2021
• <i>Technologies:</i> Python, Transfer Learning , YOLOv4, Tensorflow model Zoo, OpenCV	
Virtual Makeup Try-On System	2020 – 2021
• <i>Technologies:</i> Python, OpenCV, Deep Learning	

Technical Skills

Programming Languages: Python, C++, C, JavaScript, HTML
Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, OpenGL
Tools: Git, Latex, Anaconda, VS Code

Test Scores

Graduate Record Examinations (GRE) : Total : 307, Quant : 164, Verbal : 143

Online Courses & Certifications

• Advanced Computer Vision with TensorFlow – Coursera	[Certificate]
• Deep Learning Specialization – Coursera	[Certificate]
• Sequence Models – Coursera	[Certificate]
• Machine Learning for Everyone Track – DataCamp	[Certificate]

Awards and Honors

Best Paper Award	2021
<i>International Conference on Big Data, IoT and Machine Learning (BIM)</i>	
Dean's Award	2021 – 2022
<i>Khulna University of Engineering & Technology, Khulna, Bangladesh</i>	
Board Scholarship, Higher Secondary Certificate Examination (HSC)	2017
<i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>	
Board Scholarship, Secondary School Certificate Examination (SSC)	2015
<i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>	
Board Scholarship, Junior School Certificate Examination (JSC)	2012
<i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>	

Leadership and Affiliation

Assistant Hall Provost	2025 – Present
<i>Student Hall-2, BAUST, Khulna, Bangladesh</i>	
Senior Instructor	2025
<i>Al Biruni Society of Scientific Studies</i>	
President	2022 – 2023
<i>Jashore Association of KUET (JAK)</i>	
Senior Patrol Leader	2013 – 2015
<i>1 no Jashore Zilla School Scout Team, Jashore, Bangladesh</i>	

References

1. **Dr. Sk. Md. Masudul Ahsan**
Professor, Dept. of Computer Science & Engineering, KUET, Khulna, Bangladesh
Email: smahsan@cse.kuet.ac.bd | Phone: +880 1714 087 283
2. **Jakaria Rabbi**
Assistant Professor, Dept. of Computer Science & Engineering, KUET, Khulna, Bangladesh
Email: jakaria_rabbi@cse.kuet.ac.bd